

**The Illusion of a Monolithic Creole:
The Role of Race in the Singlish Lexicon Network**

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Author Note

Data and computational scripts for this study are available at the OSF database: [Link](#)

The video presentation for this project can be viewed here: [Video Link](#)

AI Declaration: Gemini-3 was used for debugging R code, giving feedback on drafts, refining sentence structures for clarity. I told Gemini that this is a PhD course on “Network Science in Psychology” and told it to grade it strictly. I am responsible for the ideas, content and quality of the submitted work. I acknowledge that this final report is the product of my own work and I take great pride in it. All materials consulted have been duly cited and credited.

Abstract

This study investigates whether the cognitive architecture of the Singlish lexicon is universally shared across Singaporean demographics (Chinese, Malay, Indian) or structurally mediated by race. Operating on the assumption that Singlish functions as a monolithic unifying creole, I hypothesised that overarching network topologies and anchor words would align structurally. To test this, hybrid semantic-syntactic networks were constructed, using iterative downsampling to ensure topological comparability. Results rejected the hypothesis, revealing a stratified architecture. While all demographics share a universal scaffolding anchored by English function words, the Malay and Indian anchor words largely bifurcate. These minority networks exhibited severe Jaccard fragmentation, relying on mutually exclusive, cross-pollinated loanwords. Furthermore, overarching topologies diverged: the Malay network formed a compact, assortative core, while the Indian network maintained a highly integrated web. Ultimately, associative retrieval in Singlish is a culturally mediated cognitive process.

Keywords: Singlish, mental lexicon, cross-cultural psychology, network analysis

Introduction

The mental lexicon is conceptualized as a complex network where nodes represent concepts and edges represent semantic associations, which are responses given to a presented cue word (Siew et al., 2019). The structure of semantic memory is shaped by cultural context. Cross-cultural research demonstrates that cultural background influences free verbal associations at least as strongly as language itself (Szalay & Windle, 1968), with Western and Eastern cultures differing in whether they organise concepts categorically or functionally (Leger et al., 2024). Even when comparing the same concepts across languages, core semantic features remain structurally similar but peripheral features reflect sociocultural diversity (Vivas et al., 2020).

In multicultural Singapore, the colloquial language Singlish serves as a universal bridge across ethnicities, operating as a "polyglot of English, Malay, Tamil, and various Chinese dialects" (Groppe, 2014). Recent research demonstrates that structural properties and associative pathways of the Singlish mental lexicon differ significantly from standard American English, reflecting a highly nuanced and localised cognitive structure (Chern, 2024). Psycholinguistic models of bilingual memory suggest that a speaker's primary language heavily influences how they organise secondary languages (Feng & Liu, 2024).

While Singlish acts as a shared linguistic bridge, it remains unclear whether the fundamental cognitive architecture of this lexicon is universally structured across all Singaporeans or implicitly scaffolded by the distinct ethnic mother tongues (e.g., Mandarin, Malay, Tamil) of its speakers. Therefore, this study asks: Does racial background alter the cognitive architecture of the Singlish lexicon? Since Singlish functions as a unifying national creole (Groppe, 2014), I hypothesise that demographic variations would yield no significant divergence—with global network metrics, structural alignment, and the prioritisation of central lexical anchors being statistically consistent across each racial demographic.

Methods

Data Collection and Structuring

This study uses empirical free-association data from the Small World of Singlish Words project (Wong & Siew, 2024), consisting of responses to 299 Singlish cue words collected via Qualtrics. Participants were randomly presented with Singlish cue words one at a time and tasked with providing up to three responses (R1, R2, R3). Participants could also indicate via a checkbox if they did not recognise the cue or could not provide responses.

Of the initial 610 total participants, I excluded 79 participants (12.9%) who did not meet my a priori criteria: 4 did not complete the study; 75 were not born in nor residing in Singapore. Before text processing, discrete secondary and tertiary (R2 & R3) responses were reformatted as new individual rows to form unique cue-response pairs. Integrating continuous associations (R2 & R3) alongside primary responses (R1) significantly improves the structural connectivity of the semantic network, ensuring a greater proportion of nodes are integrated into the maximal strongly connected component (De Deyne et al., 2019).

Data Processing

Data was processed similar to the Small World of Words project (De Deyne et al., 2019). First, English articles (*a*, *the*), particles (*to*), and 'self-loop' responses—where a participant simply reiterated the cue word—were removed ($N = 73$, 0.26%).

While participants were instructed to leave responses blank if unsure of a cue, there were some who responded with "*don't know*", "*not sure*" or "*no idea*". Since some cue words carry this meaning (e.g., *catch no ball*), these inputs were cross-referenced. If multiple people gave the same target phrase to a specific cue, it was kept as a legitimate colloquial association; otherwise, it was removed ($N = 62$, 0.22%).

Applying De Deyne et al. (2019) criteria on excluding participants—with more than 30% multi-word responses—would result in the loss of 50% of this dataset. Since Singlish

relies heavily on multi-word phrasal idioms, a one-word threshold is too restrictive. By adapting this to a two-word threshold instead, only 32 participants were flagged for removal (5.5% of data), successfully filtering out individuals who provided lengthy definitions rather than immediate associations. To further eliminate highly descriptive sentences while preserving valid multi-word Singlish phrases, I excluded responses with more than five words per response ($N = 193$, 0.72%). The threshold of five words was chosen to accommodate multi-word target concepts (e.g., *5Cs: cash, car, credit, condominium, club*).

The final dataset consisted of 495 participants who generated a total of 26597 valid word associations. Since this study focuses on the primary linguistic communities in Singapore, the final sample (Table 1) only involved the three major racial demographics: Chinese ($N = 474$, 95.80%), Malay ($N = 10$, 2.02%), and Indian ($N = 11$, 2.22%).

Network Construction: The Hybrid Model

Following data processing, directed network graphs were constructed for each race. In standard psycholinguistics networks, nodes represent unique words and edges represent a cue-to-target semantic association. However, since the Singlish lexicon is deeply phrasal, mapping cues to single-word targets systematically fractures idioms to represent true cognitive retrieval pathways. To resolve this, a hybrid edge-list was constructed with two strategies: **(1) Semantic Priming** (Cue $\rightarrow$ Each word in response): Semantic activation spreads to the underlying concepts of the cue simultaneously. For example, the cue *o\$ps\$* with response ‘*owe money pay money*’ now has these directed edges: *o\$ps\$* $\rightarrow$ *owe*; *o\$ps\$* $\rightarrow$ *money*; *o\$ps\$* $\rightarrow$ *pay*. **(2) Syntactic Chaining** (Word 1 $\rightarrow$ Word 2): Each word in response is sequentially related to the subsequent word in response. For example, *owe* $\rightarrow$ *money*; *money* $\rightarrow$ *pay*; *pay* $\rightarrow$ *money*. By combining unigram and sequential bigram edges, the resulting hybrid network models both the associative conceptual recall and the grammatical chaining inherent in the Singlish lexicon.

Network Analysis Strategy

After constructing the hybrid graphs, global network statistics for each race were calculated (Table 2). Due to the unequal racial representation, 100 iterative subsamplings were employed for Chinese and Malay groups at the participant level, until they roughly matched the quality of the Indian network ($N = 539$ edges). This ensured comparable network density and size across each racial group, preventing the larger Chinese sample from artificially inflating network metrics. While degree distributions in linguistic networks follow Zipf's (1949) power-law distribution, iterative downsampling allows for mean macro metrics to be calculated. By Central Limit Theorem, this sample of means approximates to a normal distribution, so Welch's Two-Sample t-tests were used to compare subsampled Chinese and Malay networks, while One-Sample t-test was employed against the Indian baseline network.

To analyse the structural hubs of the Singlish lexicon, degree centrality—defined as the number of unique edges connected to a node—was calculated for all nodes in the dataset. Degree centrality measures associative breadth and predicts lexical retrieval time, with high-degree nodes functioning as hubs for information exchange (Siew et al., 2019). To test whether each race shares a universal structural foundation, an unweighted Jaccard Similarity Coefficient was calculated. This index measured the lexical overlap between the top 20 degree-centrality hubs of each pairwise race network comparison. To test for cross-cultural structural alignment and the prioritisation of culturally specific semantic anchors, absolute degree scores were converted into normalised percentiles for a cross-demographic percentile rank-shift analysis. Spearman's rank correlation (ρ) was then calculated to evaluate whether different races rank their hubs in structurally similar or divergent orders.

Results

From Table 3, analysis of the Average Shortest Path Length (ASPL) revealed much significant structural divergence. The Malay network exhibited an exceptionally low average

ASPL ($M = 2.06$, $SD = 0.13$), which was significantly lower than the Chinese network ($M = 2.85$, $SD = 0.99$, $t(102.19) = 7.89$, $p < .001$) and the Indian baseline ($M = 3.11$, $t(99) = -82.83$, $p < .001$). Furthermore, the Chinese network's average ASPL was also significantly lower than the Indian baseline ($t(99) = -2.56$, $p = .012$). Also, the Malay network has a notably smaller average diameter ($M = 6.89$, $SD = 0.31$) than the Chinese ($M = 9.05$, $SD = 3.05$) and Indian ($M = 9.00$) networks. These results indicate that the Malay network has the most topologically efficient Singlish lexicon, with the fewest associative steps to bridge concepts.

While the Indian network has the highest ASPL, it does not indicate cognitive inefficiency; rather, it reflects a high degree of overarching global integration. The Indian baseline has the fewest number of components (51) and the largest proportion of nodes within its largest connected component (LCC) (69.8%). Therefore, while their Singlish associations require more associative steps to connect, the Indian vocabulary is far less compartmentalised into isolated micro-topic islands. Here, the Indian cognitive map forms a broader, more holistic, and far-reaching associative web.

Degree Assortativity, which measures the tendency of nodes to connect to other nodes of similar associative power, was positive for Malay ($M = .036$, $SD = .013$), while Indian ($M = -.099$) and Chinese ($M = -.004$, $SD = .047$) exhibited negative disassortative structures. In the Malay lexicon, hub words actively trigger other hub words, creating a resilient and easily accessible central vocabulary. Conversely, the negative assortativity in Indian and Chinese networks reflect a “hub-and-spoke” architecture, where central anchors connect with peripheral nodes.

Significant differences were also observed in Global Transitivity (Clustering). The Malay network showed significantly higher local clustering ($M = .26$) compared to the Chinese network ($M = .24$, $t(138.74) = -6.23$, $p < .001$) and the Indian baseline ($.24$, $t(99) = 12.37$, $p < .001$). While the Chinese network's clustering did not significantly deviate from

the Indian baseline ($t(99) = -1.19, p = .236$), the Malay network demonstrates a significantly higher propensity for semantic triangles, suggesting a more redundant and resilient local associative memory for Singlish terms within the Malay demographic.

The Illusion of a Shared Lexicon Across Races (Jaccard Similarity)

Initial analysis of the absolute highest-ranking hubs (degree centrality) revealed that while the demographic networks appear to share a baseline vocabulary, this overlap is largely an illusion driven by universal English syntax. An unweighted Jaccard Similarity comparison of the top 20 structural hubs showed that the highest overlaps occurred between the Chinese majority and the minority groups (Chinese vs Indian: *Jaccard* = .212; Chinese vs Malay: *J* = .176). A qualitative audit (Table 4) of these shared nodes revealed they consist almost entirely of standard English grammatical connectors and modifiers (e.g., *not, no, of, in*), rather than localised Singlish idioms. Crucially, Malay vs Indian networks had mostly different hub words (*J* = .081), agreeing on only three overlapping hubs (*very, old, get*). The low Jaccard Similarity scores could indicate that while all demographics utilise a shared Singlish lexicon, their localised cognitive maps are structurally distinct.

Distinctive Cultural Anchors (Degree Percentile Rank Shifts)

To isolate vocabulary items functioning as central hubs in one race while remaining peripheral in another, absolute degree scores were converted into normalised percentiles for cross-demographic comparison. This rank-shift analysis revealed cross-cultural linguistic adoption within the minority networks (Table 5). The Malays exhibited a structural reliance on Hokkien-derived idioms—such as *si mi tai dzi* (Percentile Shift = +.989), *chiong* (+.977), and *ah lian* (+.868)—as their most distinctive associative anchors compared to the Indian network. Conversely, the Indian network was anchored by Malay words, such as *atas* (Percentile Shift = -.877) and *boleh* (-.877). Here, it seems that underneath a universal

English foundation, each racial demographic utilises distinct, cross-pollinated cultural lexicons to anchor associative thought.

Structural Alignment (Spearman's Rank Correlation)

To quantify overarching structural alignment of anchor words, a Spearman's rank correlation (ρ) was conducted on degree percentiles. The Chinese network exhibited a moderately positive structural alignment with both the Malay ($\rho = .494, p < .001$) and Indian ($\rho = .520, p < .001$) networks. This suggests the Chinese network functions as the central linguistic baseline, sharing significant topological overlap with minority demographics.

However, when comparing Malay and Indian networks, the structural alignment collapsed ($\rho = -.089, p = .221$). The lack of a significant relationship shows that the ranking of cognitive hubs in the Malay network has no predictive power for the Indian network and vice versa, suggesting divergence. This reveals that Malays and Indians have different ranking orders of anchor words and different underlying architectures for lexical retrieval.

Furthermore, Figure 1 reveals demographic divergence in semantic extension—how different cultures repurpose standard English to translate colloquial concepts. The structural prominence of formal words like *acknowledge* in the Indian network, or *plan* and *able* in the Malay network, demonstrates that even when demographics utilise the exact same English vocabulary, they hold different anchor ranks and functions within their mental lexicons.

Discussion

Assuming Singlish functions as a universal, unifying creole (Groppe, 2014), this study hypothesised that racial variations would yield no significant divergence in overall network architecture. However, macro-level network metrics, degree centrality rank-shift analysis and Spearman's rank correlations fail to support this. Instead, results reveal that the shared Singlish cognitive map is a sociolinguistic illusion, masking a stratified architecture of retrieval.

The Stratified Architecture of the Singlish Lexicon

The foundational stratum is universally shared across all demographics. The statistically significant Spearman correlations between the Chinese majority and both minority groups confirm that all Singaporeans use standard English function words (*very, not, of, in, no*) as the universal functional scaffolding for retrieval. This shared layer facilitates cross-cultural communication and creates the appearance of a unified dialect. However, beneath this shared functional scaffolding lies a highly divergent content stratum. The lack of hub alignment between the Malay and Indian networks provides empirical proof of structural bifurcation. Between these networks, there is a different prioritisation of hub words, hence different immediate retrieval patterns. Besides drawing from a universal pool, both groups anchor their Singlish lexicon with distinct culturally specific content words.

Beyond specific lexical anchors, the overarching network architectures also diverge significantly. The Malay network has a structurally cohesive core, is tightly clustered, positively assortative and possesses a small diameter in LCC, making it highly efficient for rapid retrieval of common Singlish terms. Conversely, the Indian network operates as a wider associative net. While disassortative and having higher ASPL—being more spread out—the Indian network is more integrated, featuring fewer components and a larger LCC. This allows the Indian demographic to capture a vast amount of vocabulary within a single, overarching cognitive web, while the Chinese network functions as a topological median between the two.

Lexical Cross-Pollination and Extension

Crucially, the distinct cognitive maps of the minority demographics are not simply isolated islands of their respective mother tongues. The percentile rank-shift analysis revealed a profound degree of lexical cross-pollination. The Indian network is heavily anchored on Malay loanwords (*atas, boleh*), while the Malay network relies on Hokkien-derived phrases (*si mi tai dzi, ah lian*). This suggests that minority groups actively adopt and structurally

prioritise the vernacular of other groups to anchor the Singlish lexicon, rather than defaulting to their own mother tongue. The network data demonstrates that the demographics are not merely borrowing superficial lexicals. Rather, they are anchoring their cognitive maps on the localised loanwords of out-group communities, a phenomenon that aligns with deep structural cross-linguistic influence in Singlish (Teo, 2020).

Network Resilience and Targeted Cognitive Fragmentation

The structural divergence of these cultural hubs implicates the cognitive resilience of the Singlish lexicon. Since all races share similar functional English words as hub words, a generalised network attack on highest-degree nodes could impact all groups equally.

However, the vulnerability of each network is also culturally relative—rooted in their distinctive hub words. If one were to theoretically simulate a targeted attack to eradicate the Singlish lexicon, the specific nodes required to fracture the network would differ by race.

Deleting uniquely prominent Hokkien hubs would fragment the distinct associative pathways of the Malay network, whereas removing specific Malay loanwords would disproportionately cripple the retrieval efficiency of the Indian network. This demonstrates that beyond the shared English skeleton, some structural hubs of the creole are uniquely adapted to each race.

Limitations and Methodological Considerations

Despite the divergence identified, these findings should be interpreted with caution due to size imbalances. The original dataset collected aimed for each cue to be responded by at least 100 participants to ensure sufficient quality (Wong & Siew, 2024). However, most of these respondents were Chinese, limiting the appropriateness of this dataset in investigating cross-cultural differences. The Chinese group ($N = 474$) was robust enough to achieve high lexical saturation, providing a highly stable topological map. In contrast, the Malay ($N = 10$) and Indian ($N = 11$) networks are susceptible to data sparsity. Here, the Chinese network being more robust could explain the greater structural alignment with minority networks.

Specifically, the most extreme percentile shifts observed between Malay and Indian networks were driven by nodes with one degree in their respective groups. While this may represent mutually exclusive cultural concepts, it is also probable that these shifts are data artifacts. Small sample sizes struggle to capture the long-tail distribution of a vocabulary, meaning “missing” edges may be unobserved rather than culturally non-existent, especially if participants lacked exposure to certain cue words. Additionally, these distinctive anchors may be confounded by word frequency effects. Singlish terms with high ubiquitousness within a group naturally elicit wider variance of free associations, inherently inflating degree centrality. Therefore, future research must prioritise balanced, large-scale sampling of minority races to ensure full lexical saturation to confirm whether the identified hub words remain mutually exclusive at scale.

Additionally, the decision to analyse the semantic-syntactic networks using unweighted degree centrality introduces a systematic bias regarding lexical diversity. Because unweighted degree prioritises associative breadth over consensus, cue words that elicited highly verbose or idiosyncratic explanations naturally accumulated higher out-degrees, artificially inflating their structural rank compared to high-consensus, single-word responses. While a strict five-word maximum threshold was implemented to contain artificial inflation, absolute degree rankings in this study should be interpreted as a measure of semantic variance rather than frequency of use.

Ultimately, this report contributes a network science perspective to the study of creole languages. Mapping the Singlish mental lexicon, I challenged that Singlish is a monolithic national creole, proving that race mediates associative memory and lexical retrieval. Multilingual populations construct functionally stratified architectures, relying on shared syntax for communication while using some cross-pollinated hubs for semantic anchoring.

References

- Chern, J. J. S. (2024). *Comparing cultural differences between speakers of different English dialects through the perspective of semantic networks and associative learning* [Master's thesis, National University of Singapore]. ScholarBank@NUS.
- De Deyne, S., Navarro, D. J., Perfors, A., Brysbaert, M., & Storms, G. (2019). The “Small World of Words” English word association norms for over 12,000 cue words. *Behavior Research Methods*, 51, 987–1006.
- Feng, X., & Liu, J. (2024). The structure of lexical-semantic networks at global and local levels: A comparison between L1 and L2. *Complexity*, 2024, Article 8644384. <https://doi.org/10.1155/2024/8644384>
- Groppe, A. M. (2014). 'Singlish' and the Sinophone: Nonstandard (Chinese/English) languages in recent Singaporean cinema. In A. M. Groppe & M. Hillenbrand (Eds.), *Sinophone cinemas* (pp. 147-168). Palgrave Macmillan. https://doi.org/10.1057/9781137311207_9
- Leger, K. R., Snyder, H. R., Boduroglu, A., & Gutchess, A. (2024). The role of culture and semantic organization in working memory updating. *Memory*, 32(7), 576-586. <https://doi.org/10.1080/09658211.2024.2351062>
- Siew, C. S. Q., Wulff, D. U., Beckage, N. M., & Kenett, Y. N. (2019). Cognitive network science: A review of research on cognition through the lens of network representations, processes, and dynamics. *Complexity*, 2019, Article 2108423. <https://doi.org/10.1155/2019/2108423>
- Szalay, L. B., & Windle, C. (1968). Relative influence of linguistic versus cultural factors on free verbal associations. *Psychological Reports*, 22(1), 43-51. <https://doi.org/10.2466/pr0.1968.22.1.43>

- Teo, M. C. (2020). *Crosslinguistic influence in Singapore English: Linguistic and social aspects* (1st ed.). Routledge. <https://doi.org/10.4324/9780429463853>
- Vivas, J., Kogan, B., Romanelli, S., Lizarralde, F., & Corda, L. (2020). A cross-linguistic comparison of Spanish and English semantic norms: Looking at core features. *Applied Psycholinguistics*, 41(2), 285-302. <https://doi.org/10.1017/S0142716419000523>
- Wong, J. J., & Siew, C. S. Q. (2024). Preliminary data from the Small World of Singlish Words project: Examining responses to common Singlish words. *Journal of Open Psychology Data*, 12(1), 3. <https://doi.org/10.5334/jopd.108>
- Zipf, G. K. (1949). *Human behavior and the principle of least effort: An introduction to human ecology*. Hafner.

Table 1*Demographic Breakdown of Participants and Valid Semantic Responses*

Race	No. of Participants	(%)	Valid Responses	Avg. Responses per Participant	Total Nodes	Total Edges
Chinese	474	95.8	25678	54.2	6748	10938
Malay	10	2.02	538	53.8	619	488
Indian	11	2.22	381	34.6	473	353

Table 2*Macro Network Statistics of Each Raw Demographic Network*

Network Statistics	Chinese	Malay	Indian
Nodes	4345	653	539
Edges	17315	826	687
Average Degree	7.97	2.53	2.55
No. of Components	1	70	51
LCC Size	4345	448	376
Proportion of Nodes in LCC	1.00	.69	.70
ASPL	7.48	2.22	3.11
Transitivity	.06	.24	.24
Diameter	21	7	9
Edge Density	.0009	.0019	.0024
Assortativity (Degree)	-.081	.041	-.099

Note. LCC = Largest Connected Component; ASPL = Average Shortest Path Length

Table 3*Macro Network Statistics of Each Subsampled Demographic Network*

Network Statistics	Chinese Subsampled		Malay Subsampled		Indian (Baseline)
	Mean	SD	Mean	SD	
Nodes	601.6	35.0	597.9	18.4	539
Edges	733.7	33.8	726.8	25.4	687
Average Degree	2.44	0.12	2.43	0.07	2.55
No. of Components	72.3	10.2	72.7	5.9	51
LCC Size	349.6	43.5	381.8	22.1	376
Proportion of Nodes in LCC	.58	.07	.64	.03	.70
ASPL	2.85	0.99	2.06	0.13	3.11
Transitivity	.24	.02	.26	.01	.24
Diameter	9.05	3.05	6.89	0.31	9
Edge Density	.0020	.0002	.0021	.0001	.0024
Assortativity (Degree)	-.004	.047	.036	.013	-.099

Note. “Subsampled” represents the average metrics across 100 iterations of random downsampling to match the Indian network baseline. LCC = Largest Connected Component; ASPL = Average Shortest Path Length

Table 4*Cross-Cultural Jaccard Similarity Scores for Top Degree Centrality Words*

Comparison	Jaccard Score	Shared Count	Shared Words
Chinese vs Malay	.176	6	not, no, very, on, good, what
Chinese vs Indian	.212	7	of, in, very, up, something, it, bad
Malay vs Indian	.081	3	very, old, get

Table 5*Top 15 Distinctive Lexical Anchors Between Malay and Indian Networks*

Word	Malay Network		Indian Network		Percentile Shift	Dominant Group
	Degree	Percentile	Degree	Percentile		
rabak	10	.989	1	0	.989	Malay
si mi tai dzi	10	.989	1	0	.989	Malay
chiong	8	.977	1	0	.977	Malay
kayu	8	.977	1	0	.977	Malay
eye power	7	.957	1	0	.957	Malay
go	7	.957	1	0	.957	Malay
do	6	.928	1	0	.928	Malay
rabz	6	.928	1	0	.928	Malay
shit	1	0	6	.920	-.920	Indian
atas	1	0	5	.877	-.877	Indian
boleh	1	0	5	.877	-.877	Indian
gahmen	1	0	5	.877	-.877	Indian
ord	1	0	5	.877	-.877	Indian
ah lian	5	.868	1	0	.868	Malay
lobang	5	.868	1	0	.868	Malay

Figure 1*Cognitive Hub (Anchors) Alignments across Demographics*